SpriteFusion 2D

From generated art to game-ready pixel assets — on AMD Radeon.

Track 1: Multimodal Content Creation Tools | Team: kaggle4youAMD

21 ms
PER SPRITE ON ROCm
512×512 palette + grid recovery on AMD Radeon (`gfx1100`), ROCm 7.2 — 29× the same host's CPU, receipted & reproducible

A deterministic finish for AI-generated game art

A diffusion model gives you a picture. It does not give you a clean pixel grid, a small reusable palette, a transparent silhouette, or a history you can roll back. SpriteFusion 2D is the desktop workshop that turns that picture into an asset your engine can actually import — end to end, on Radeon hardware, with a project/revision timeline instead of a pile of loose PNGs.

Why the number is real

- Every run emits a versioned JSON *receipt*: command, environment, thread count, SHA-256 of source/input/output, raw timings, git state.
- 512×512, 32-color palette: host CPU median **0.6157 s** → AMD Radeon ROCm median **0.0211 s** — **29.2**×, cross-checked at 28.4× on a second config.
- Baseline stated, not hidden: EPYC 9334 on a 16-CPU quota with PyTorch threads pinned to match. Against a fast laptop CPU the same result is ∼8× — both numbers are in the profile.
- Same `torch.cuda` call path as CPU/NVIDIA — zero AMD-specific branches, validated on `gfx1100` / ROCm 7.2 / PyTorch 2.9.1.

Pipeline



What's inside

Background removal
Transparent-silhouette extraction, ready for outline and packing steps.

Outline generation
Configurable outline pass for sprite readability at game resolution.

Pixel-art conversion & palette recovery
Device-native k-means palette learning and grid recovery, tiled for large sheets, with a golden-image exactness suite.

ComfyUI generation, in-app
Submit, watch progress, and commit a chosen candidate straight into the project — no manual file shuffling.

Full revision history
Project-based undo/redo; every processing step is a traceable, revertible revision.

Validated, not just claimed

377	99	gfx1100	0
Python backend tests	frontend tests	AMD Radeon target, ROCm 7.2	fabricated numbers — every figure ships with a JSON receipt